

Robust low-cost passive UHF RFID based smart shopping trolley

Tharindu Athauda, Juan Carlos Lugo Marin, Jonathan Lee, Nemai Karmakar

Department of Electrical and Computer Systems Engineering
Monash University, Clayton, Victoria 3800, Australia

Tharindu.Athauda@monash.edu, Nemai.Karmakar@monash.edu,
jclug1@student.monash.edu, jwlee51@student.monash.edu

Abstract— Retailers are often interested in low cost mechanisms to maintain stocks as well as for tracing products across the supply chain in an efficient and effective manner. In addition, shoplifting is another concern face because of the lack of effectiveness in product tracing technique such as “barcode” used in retail supermarkets. “AmazonGo” a smart retail layout which was introduced by Amazon, to address above issues was found to be inefficient due to the over dependency of system based on historical purchased patterns of consumers. In this study, we propose a low-cost, robust, passive UHF RFID based shopping trolley system which allows tracing and processing shopping data in real time. The UHF antenna mounted shopping trolleys are defined “Smart Trolleys” while shopping items are tagged using UHF RFID tags with unique identification codes.

Keywords—Smart Shopping Trolley, UHF RFID, RSSI, Circular Polarised Antenna, Branch Line Coupler.

I. INTRODUCTION

People used to have a list of items written in a piece of paper when they went shopping for groceries; however, the advancement of technology has changed how people do shopping over the last decade [1]. In addition, the emergence of smart phone has changed the retail shopping experience drastically. The retailers are continuously working on improving the shopping experience to make sure that their customers are satisfied in overall shopping experience.

There have been various attempts which have been carried out to eliminate lengthy shopping lines in retail stores. One of the famous approaches is the introduction of self-checkouts where customer convenience has been improved drastically [2]. Self-check outs have been popular since then due to low overhead cost; however, the shoplifting and lower operating efficiencies are considered as major drawbacks in the retail environment.

The selection of a single technology that can be used to enhance shopping experience is a difficult task as the

expectation in a brand clothing store to grocery store can be quite different, and can also depend on individual perception [3]. In addition, it has been found that consumers prefer to have item level information to make purchase decisions [4]. Moreover, demands for visual technology and privacy have a greater influence on consumer satisfaction. Therefore, it is important to identify ways to improve shopping experience while the factors such as return on investment, expected sales growth and meeting customer expectations are also considered.

There have been three major attempts, where shopping trolley was used as the medium to improve shopping experience. In 2005, Fujitsu, a Japanese company has demonstrated a shopping trolley with an inbuilt barcode scanner. The barcode scanner was used to scan both products and loyalty cards in real time [5]. However, this shopping trolley does not fully solve problems such as those in store stock management and shoplifting. Amazon has recently come up with a smart retail store concept “AmazonGo”[6] where the customer pick a product from the product gets tracked and self-checked. The AmazonGo system uses image processing, neural networks and deep learning algorithms to forecast which item is picked by the consumers. The system accuracy largely depends on consumers historical purchased patterns. AmazonGo also uses sensor fusion techniques to process multiple images taken from cameras around the retail store to predict best estimate of an item picked by consumers in real time [7]. Finally, Panasonic, another Japanese company has come up with a RFID based shopping basket where each item is tagged using UHF RFID (ranging from 916-924MHz) [8]. Panasonic has revealed that this smart basket is part of “cashier free convenient store concept” where customers can drop items into the basket and each product gets scanned through a self-serving kiosk. A higher capital cost as well as the limited size of the basket have been identified as the major drawbacks of this implementation [9].



Figure 1 - Self Registration Machine Console (a) Shopping Basket
(b) RFID integrated shopping console

It is clearly evident from the above analysis that shopping trolley is considered preferable over self-checkout systems. It also concludes that there is a clear need for a low cost and robust technology implementation which is efficient and highly accurate when dealing with shopping information. Therefore, this work aims to design and implement UHF RFID based smart trolley which can track products and provide item level information to consumers in real time.

II. SYSTEM DESIGN

UHF RFID was renowned for traceability applications and it was introduced to the mainstream by Wal-Mart [10]. Today UHF RFID continues to thrive in retail applications and fit well with shopping trolley application due to the following attributes.

- Non-line-of sight reading ability - In the shopping trolley items can be read without having to maintain a clear line of sight.
- Reading distance - UHF RFID tags can be used in both near field or far field applications which is useful in reading items close to the antenna and also from a considerable distance.
- Higher data rates – which is very useful in storing more data efficiently.
- Tag readability - UHF RFID tags can read multiple tags at one time (300 tags/seconds), which increases reading efficiency.
- UHF RFID tags are globally standardized under the ISO 18000 in particular and GS1 standards on EPC Global based on its application. Other standards such as GATG, and CEPT are also found to be relevant in industry deployment.

The above has been taken into consideration when drawing the requirement specification. The following requirements are taken into consideration in designing the smart trolley application.

- **Every item that goes in and out of the shopping trolley should be detected** – In such a case, antennas should be placed in strategic positions where the radiation patterns are able to cover the entire area inside the trolley irrespective of tag orientation.
- **Capability of UHF RFID Reader** – UHF RFID reader should be able to deal with multiple antennas with different configurations simultaneously.

It is essential to understand how UHF RFID system works in order to find solutions that cater to above stated requirements. Every UHF RFID tag has its own identity and it also consists of a RF trans-receiver. RFID reader transmits the encoded radio signal to check the presence of UHF RFID tag using the transmitter antenna. In the event that the UHF tag is within the reading range of the transmitter antenna, the tag will receive the transmitted signal. Once the tag receives the signal, the induced power in the tag antenna will activate the IC on the tag by the induced current on tag antenna and then the signal backscatters to the receiver antenna with the information retrieved from the Tag IC. Thereafter, the receiver antenna receives the tag information and send it to the RFID reader for

further processing. In this application, it is important to consider both near field and far field patterns in antenna development as both regions will be used for the tag detection. More detail about field analysis will be explained in the implementation section.

UHF RFID tags typically operate from 860-960MHz and each country has its own UHF bands. Specifically, for Australian standards, the RFID spectrum is 918-926MHz at 1W EIRP (Equivalent Isotropically Radiated Power) or 920-926MHz at 4W EIRP.

The main aim of this research is to develop a UHF RFID system that works simultaneously with multiple antennas inside a shopping trolley. The RFID reader can connect with the interface and show real time information to consumers. Therefore, significant effort is needed to design the required antennas and its optimization, to implement hardware configuration with RFID reader (CSL468 UHF RFID reader from Convergence Systems Limited), to validate system using UHF RFID tags and to develop the user interfaces.

The necessity to read RFID tags in different orientations is considered essential; hence, the circular polarized (CP) patch antenna is selected. CSL468 RFID reader is selected as it provided multiple antenna ports as part of system configuration. A low cost RFID reader can be suggested later, to gain the commercial advantage in mass deployment of this application.



Figure 2 – Convergence Systems Limited – CSL468 RFID Reader

Table 1 –CSL468 RFID Reader properties	
Property	Value
Number of ports	16
Tagging Speed	300 tags / second
Read Range	12meters

Considering the above research goals and requirements, the following smart trolley design is suggested.

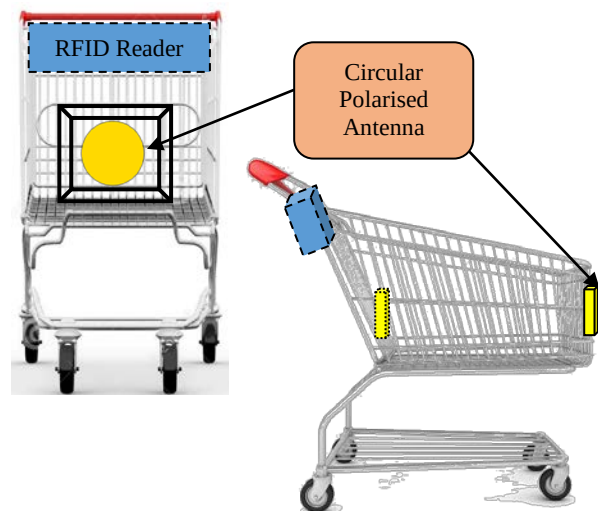


Figure 3 – Systematic (a) Front view (b) Side view with Circular Polarised Antennas of the Smart Shopping Trolley

The next step is to understand the directional gains needed for the application of the CP antennas in order to obtain RSSI (Return Signal Strength Indicator) values for the application.

III. SYSTEM DEVELOPMENT

As the passive UHF RFID tags are not inbuilt with a power source, the readability of the tag is purely based on the transmitted power from UHF RFID reader through the transmission antenna to the tag. Such a signal in real time application travels two times along the path of the signal. Friss has explained this scenario by considering the power received by the UHF RFID reader (P_R) once it is transmitted and backscattered by the UHF RFID passive tag as below [11].

$$P_R = P_T G_T^2 G_R^2 \zeta \left(\frac{\lambda}{4\pi R} \right)^{2\eta}$$

where P_T is the power transmitted, G_T and G_R are directional gains of the k^{th} tag and reader antenna, respectively, considering the direction of signal propagation. ζ denotes the efficiency of backscattered transmissions of the tag (which is in general considered as -5dB [12]), R denotes the range between the tag and antenna at a given instant and λ is for the wavelengths of the transmitted signal. η is heavily dependent on propagation environment; however, $\eta = 2$ has been considered for this application. This value is derived, considering the line of sight nature between the RFID tag and the reader. Therefore, RSSI (Return Signal Strength Indicator) measurement can be deduced as follows.

$$RSSI = -20 \log D + T_x \text{Power}$$

RSSI measurement plays a significant role in this smart trolley application as RSSI measurement indicates the directional gains that are needed for the antenna development. Furthermore, the CSL486 RFID reader is capable of providing derived measurement such as the “Geiger count” which indicates the average RSSI value as a % of transmitted signal power. A higher number will stand for a closer tag reading [13] and “0” will represent that the tag is not reachable at all. Geiger count can also be used to change transmitted power on demand, in order to increase the chances of tag readability.

RSSI is known to be heavily dependent on distance in the state of the art, it is also known that RSSI is dependent on environmental conditions. Metal objects, liquids and human interference can greatly alter the RSSI measurement. RSSI value can also be dependent on the tag orientation [14]. In smart trolley application, antenna should only read items inside the trolley at a given time. On the other hand, the beam pattern of the antenna should cover the full internal area of the trolley, in order to reach each tag attached to the products. Furthermore, it was decided to develop an antenna system that can cover the full range of UHF; therefore, it can cover global UHF standards (850 - 960 MHz) [15] enabling the use of developed system implementation across any country without a limitation.

CP antenna has a shorter reading range compared to linear polarized (LP) antenna. This happens due to the loss of power around 3dB at the coupler (Power get divided due to the split of lines). Both LP and CP antennas provide similar gains even though the CP is capable of reading tags in different orientations. Therefore, there is a need to develop a CP antenna as well as the hybrid coupler which aids CP antenna to provide circular polarized behavior. Considering the directional gains from UHF RFID tags, the antenna gain can be calculated for a given wavelength. In order to make the antenna design simpler and cost effective, the below circular patch was selected, as shown below.

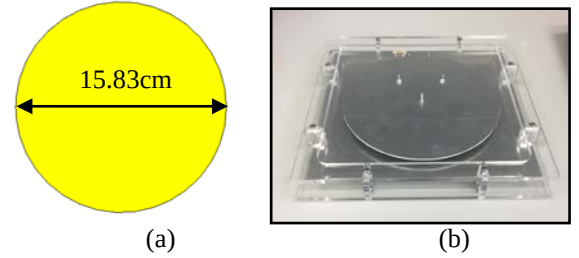


Figure 4 – Circular polarised patch antenna (a) Simulated design (CST Microwave studio 2016) (b) Fabricated patch antenna

The simplest form of coupler, the branch line coupler was selected for the application. Each transmission line is a quarter wavelength; however, this may result in reduced bandwidth which may adversely affect system requirements [16]. As shown in **Figure 5**, branch line coupler has 4 ports for input, output, coupled and for isolation. 900MHz has been selected as the center frequency based on the required bandwidth and wavelength which was calculated accordingly. FR4 has been selected to fabricate the coupler; hence, it is essential to calculate effective wavelengths as below.

$$\lambda_0(m) = \frac{c \left(\frac{m}{s} \right)}{f(Hz)} = \frac{3 \times 10^8}{900 \times 10^6} = 0.333 m$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_r}} = \frac{0.333}{\sqrt{4.6}} = 0.155 m$$

$$\frac{\lambda_g}{4} = 38.854 mm$$

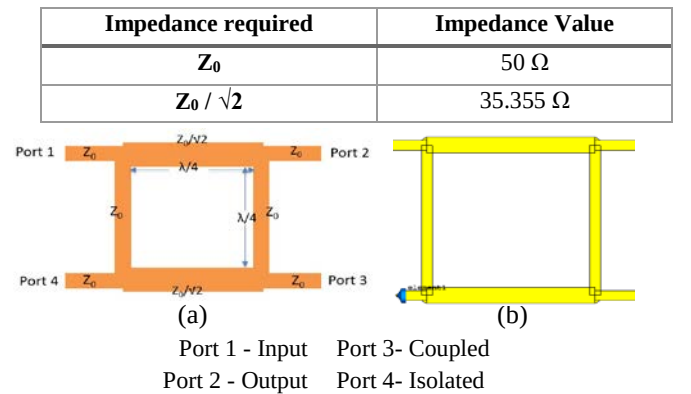


Figure 5 – Branch line coupler (a) Schematic (b) CST Microwave Studio 2016 Simulation

A characteristic impedance of 50Ω was used, as the cables and connectors had an impedance of 50Ω , and the corresponding micro strip widths were calculated using ADS Linecalc [17]. The coupler dimensions mention in **Table 1**, was used to design the coupler. Thereafter, coupler was simulated to measure values of return loss (S11), S21 and S31. In order to have a successful working simulation, S11 should be less than -10dB and $S21 = S31 = -3\text{dB}$ as power gets divided equally between ports 2 and 3. Moreover, the difference between phases of ports 2 and 3 has to be 270 degrees (out of phase by 90 degrees) in order to get ideal circular polarization where two signals from ports 2 and 3 are perpendicular to each other.

Table 1 – Coupler Dimensions	
Variable	Dimension
LAM4	46.500 mm
L50	11.511 mm
W50	3.103 mm
W35	4.992 mm
Substrate thickness	1mm
Substrate length	81.795mm
Substrate width	69.522mm
Conductor thickness	17 μm

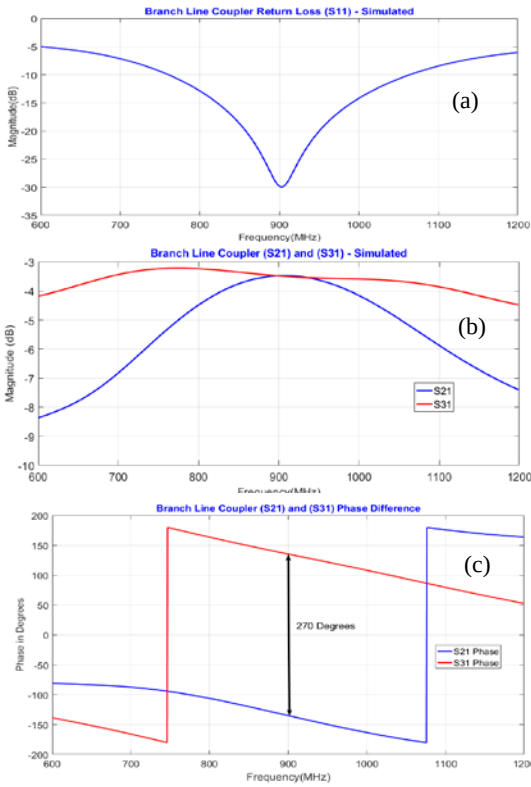


Figure 6 – Branch line coupler (a) Return loss (S11) (b) S21 and S31 (C) Phase difference

It was observed that the outcomes of S11/S21/S31 were within the acceptable ranges in power (magnitude in dB). In addition, the phase difference of 270° was observed as the required phase change to propel circular polarized properties. Therefore, the measurements of return loss of CP antenna and coupler together were taken as shown in **Figure 7**. The return loss measurement was used to validate that the performance of CP antenna and coupler together, to achieve the required bandwidth of 850-960MHz. Thereafter, both CP antenna and coupler were fabricated, in order to measure its performances for the smart trolley application.

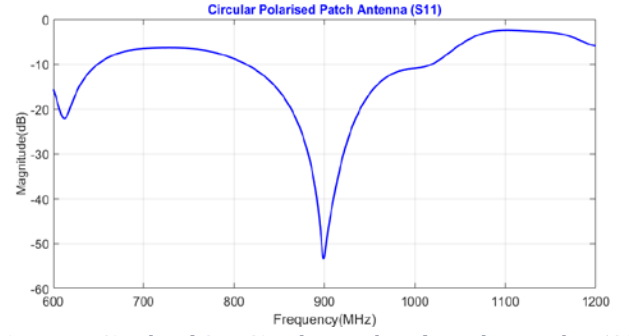


Figure 7 – Simulated S11 Circular patch and coupler together (CST Microwave Studio)

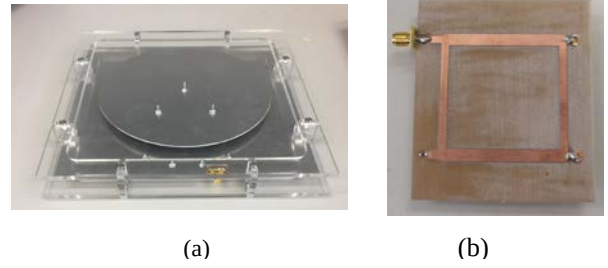


Figure 8 – (a) Fabricated Circular Patch Antenna (b) Fabricated branch line coupler (Top View)

Finally, the fabricated outcomes were compared against the simulation results as shown in **Figure 9**. Vector Network Analyzer (VNA) was used to measure the actual performance once the antenna and coupler were integrated to perform as one system.

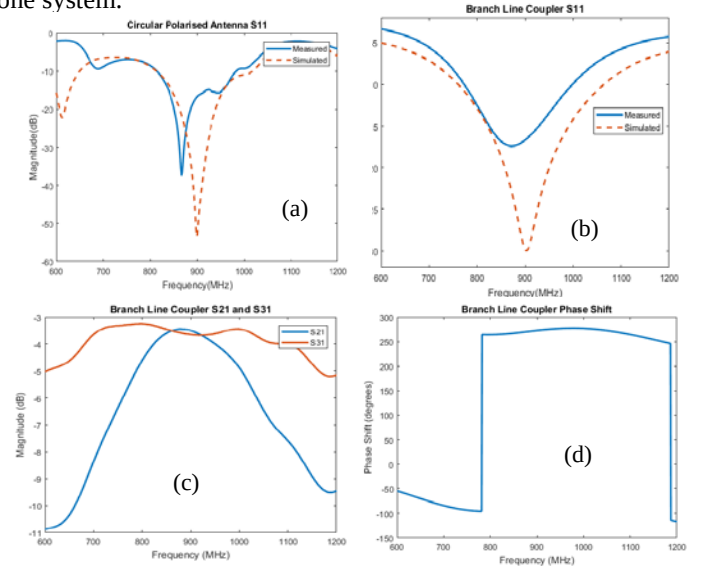


Figure 9: (a) S11 CP antenna simulated vs measured performance (b) Coupler S11 simulated vs measured (c) Coupler S21, S31 simulated vs measured (d) Coupler S21 & S31 phase difference measured

As shown in **Figure 9 (a)**, the measured S11 of CP antenna was shifted to the left from the simulated result. In the coupler, measured S11 was observed and it provides a lower amplitude (return loss of -17.35dB) and a frequency shift (870MHz) compared to simulated S11 at 900MHz. The measured S21 and S31 of the coupler were recorded as -3.46 and -3.65dB at 900MHz (center frequency) respectively. The lossy nature of FR4, could be the reason behind such variations of measured results in contrast to simulation. Finally, the phase difference

between ports 2 and 3 was found to be 272.74 degrees, where the slight mismatch in magnitude and phase represent the non-perfect behaviour of the CP antenna system. Nevertheless, for the UHF RFID tag reading purpose, these differences can be negligible.

IV. IMPLEMENTATION AND CONFIGURATION

As for the next step, the fabricated CP antenna and coupler were mounted to the shopping trolley as shown below.



Figure 10 – CP antenna attached to the shopping trolley

The positioning of antennas were positioned in a way that, it maximizes the area of coverage considering the gain and directivity of the CP antenna. The effectiveness of such placements in both near field and far field can be elaborated using E field and directivity configuration at 900MHz as shown below.

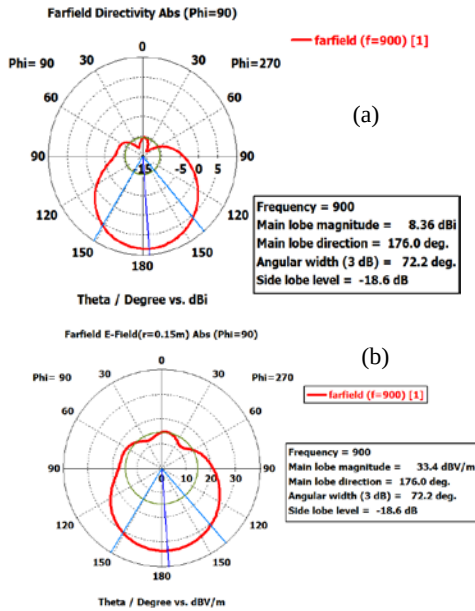


Figure 11 – (a) Far field directivity pattern of CP antenna (b) Near field E field pattern of CP antenna

As explained in the **Figure 11 (a)**, the main lobe covers a significant area in front of the CP antenna; however, some products may not be reachable by the antenna pattern especially in the corners of the trolley as shown in **Figure 12**. Such a scenario can be explained using the near field pattern (at 15cm based on $2D^2/\lambda$) as in **Figure 11 (b)**. In near field, CP antenna has a greater field intensity in both longitude and the plane parallel to the antenna. This understanding helps to explain the CP antenna's capability to detect tags at a closer proximity as well as in a distance.



Figure 12 – Product testing with smart shopping trolley

Thereafter, the item level tracking test was carried out by tagging individual products using UHF RFID tags where each tag has a unique EPC (Electronic Product Code) code. The UHF RFID tags from Robot gear, Australia [18] have been used for this experiment which has following properties.

- ISO / IEC 18000-6c
- EPC Global Gen 2
- 512 User bits and 800 Bits of memory
- 64 Bit unalterable serial number as unique ID

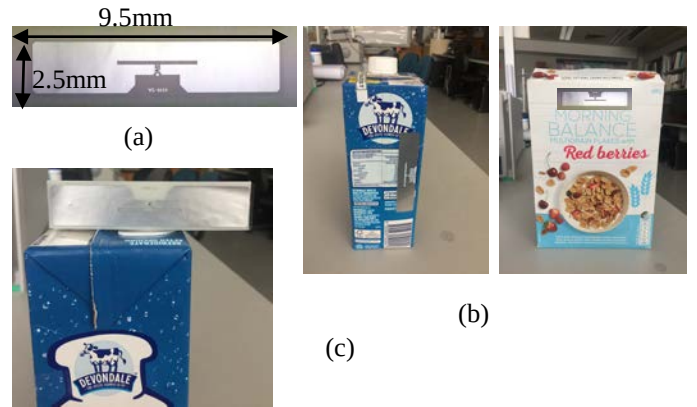


Figure 13: (a) UHF RFID Tag from Robot Gear Australia (b) Tagged products with UHF RFID tag (c) Augmented placements to support read range

Interestingly, there are instances where UHF RFID tags show poor reading performances in the presence of metal packaging such as canned foods and liquids. Augmented placements were implemented using 3D printed (using relatively lower cost ABS (acrylonitrile butadiene styrene)) frames and use those frames as a base to sticker UHF RFID tags to avoid a problematic reading zone as shown in **Figure 13 (c)**. However, this augmented carrier is not a cost viable solution in mass deployment; hence, the development of a flexible folded dipole UHF RFID tag with a ground plane is necessary to get rid of a separate carrier. For example, Xerafy Mercury metal skin inlay UHF RFID label [19] may be considered as an alternative UHF RFID tag that can be used for item-level tagging. In Xerafy tags, an aluminum foil is wrapped around (to connect both antenna and ground) a thin plastic sheet which acts as a dielectric substrate.



Figure 14 – Xerafy metal skin inlay tag

The alternative benefits brought by this tag is, its alignment with EPC Class 1 Gen 2; ISO18000-6C and works from UHF 860-960MHz covering the global range of UHF RFID. The main concern of such a tag is the cost compared to the non-specialized UHF RFID; however, smaller size, volume in mass deployment will reduce the unit cost drastically.

In the implementation process, it was decided to use existing reader hardware as explained in **Figure 2** as well as to use reader software to verify initial readings of the UHF RFID tags with the products. Each UHF RFID tag has given an EPC code where each tag ID can be reprogrammed to get detailed information such as product name, expiry date, batch number, manufacture information etc. In practice, RFID reader gets directly connected to the PC via a standard network cable and transferred data such as RSSI readings use for analysis of gathered information.

As the final experiment, the detecting of tags from various locations inside the trolley has been performed. This experiment was conducted to validate the fabricated CP antenna and coupler performances inside the smart shopping trolley. The software which was inbuilt to CSL CS-468 reader used to determine the average RSSI values as a % of transmitted power using the Geiger count.

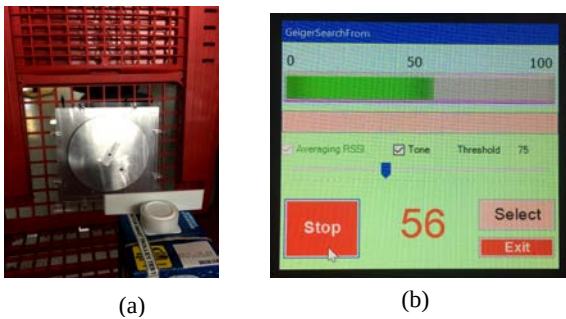


Figure 15 – (a) Experiment setup to measure Signal Quality (b) Average RSSI value as a measurement

The gathered values of RSSI, have been plotted to show the tag detection space inside and outside the trolley using a 3D space modeling as shown below. It is important to consider that these values were obtained from a single CP antenna. It is useful to identify the coverage produced by each antenna and then evaluate performances of combined antennas in the trolley application as shown in **Figure 16** below.

- Green circle - Average RSSI over 50
- Yellow circle - $40 < \text{Average RSSI} < 50$
- Red circle - Average RSSI < 40

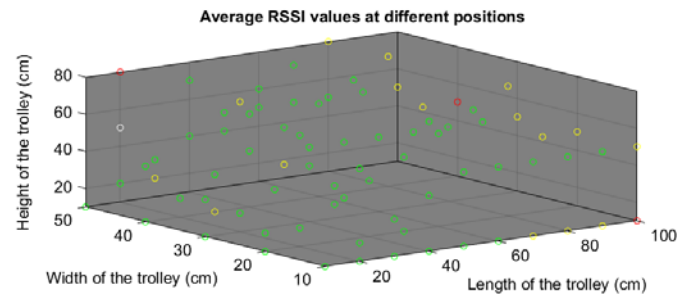
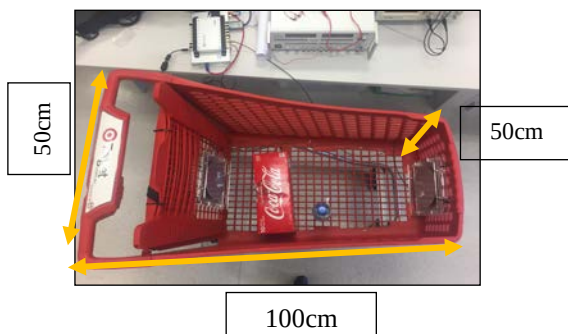


Figure 16 – Geiger count % vs Distance measurement using singular circular polarised antenna as compared to trolley measurements

It is quite evident that even a single antenna can produce a significant coverage inside the trolley; hence, it can be concluded that both CP antennas together can produce a better coverage. The final experiment was carried out, to verify the effectiveness of tag readings in presence of metal frames. In this experiment, metal skin UHF RFID tag used with metal products and the following observations are recorded.

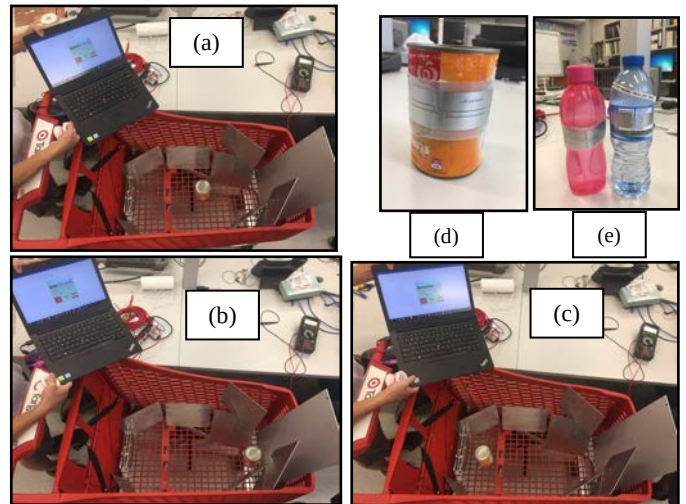


Figure 17 – (a) Reading of canned food with xerafy tag @ 50cm (b) @ 90cm (c) @ 12cm from CP antenna (d) Canned food (e) water bottles tagged with Xerafy metal skin tag

As seen in the figure above, xerafy tagged (aluminum ground plane with dielectric substrate) can be easily detected in a closer proximity. The Geiger counter that shows average RSSI value gets higher when the object is closer to the antenna and reduced when it's far away. The use of metal frames has not significantly impacted the RSSI value (when compared with **Figure 16 (b)**). The experiment has further validated that the tag reading is quite possible in presence of metal and liquid. The readings which were observed from the outside (very close to the antenna) of the trolley has a lower average RSSI value (usually less than 30).

The over the counter software inbuilt to the RFID reader also has the functionality to detect multiple tags at the same time as long as the antenna can read the tags with over 30% Geiger count (Average RSSI) as a %. These Geiger % value which determines the detectability of a given tag quality can be rearranged in the software application; however, it was decided to maintain a higher Geiger % in order to discourage readings coming from outside the shopping trolley (eg: Tag IDs from an adjacent trolley).

Finally, the user interface was developed which helps to provide real time information to consumers.

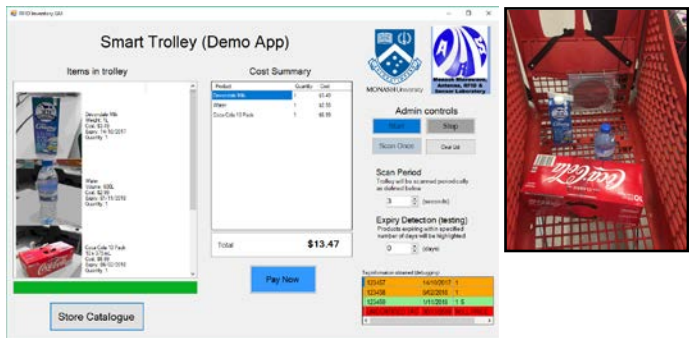


Figure 18 – Shopping trolley interface

The interface provides real time information to the consumer while the app gets refreshed at every 2 seconds to capture new information. The item level price / total and expiry dates of each item are shown on the interface and the interface will link the trolley to the payment gateway for payment processing.

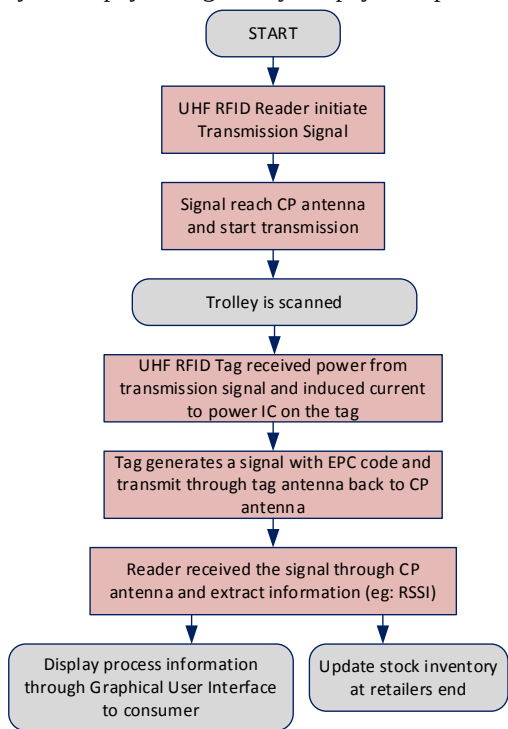


Figure 19 – Process flow chart of the system

The above flow chart can be used to summarize the complete working cycle of the UHF RFID smart shopping trolley system. In this flow diagram, critical activities are taken into account, which needs careful consideration in practical implementation.

V. COST ANALYSIS

An approximate cost analysis has also been conducted at the end of the development. The approximated cost was calculated considering the market demand (eg: approximate number of shopping trolleys in Australia), hardware cost (UHF RFID reader, antennas, jigs, tags, augmented placement carriers),

Repetitive use [Replace annually], Shopping trolley shrinkage [10% per store] and the software platform.

According to the investigation from web sources, it was revealed that Australia [20] has over 650,000 shopping trolleys across multiple super market (over 3500 outlets) chains where each store has approximately 200-350 shopping trolleys. Above approximations are considered along with the 10% shrinkage for the following calculations.

Table 2 – Cost Analysis			
Hardware cost (Approximate)		Volume	AUD \$
UHF RFID reader with 2 antenna ports [21] for 650,000		1	\$250
UHF RFID tag [considering the average of 9000 - 47000 products per grocery store [22]]		1	\$0.05
Aluminum Antenna + Laser perforated plastic jigs [02 antenna per each trolley]		2	\$28
Augmented Placement carrier		1	\$0.2
Powering the reader system using an rechargeable integrated battery		1	\$150
Bluetooth adapter to communicate with kiosk		1	\$6
Localization receiver (Installed in super market and depends on the size of the super market) – Not considered in calculation		10-30	\$100-200
Software Cost (Approximate)			
Reader platform (Inbuilt)			No cost
Configuration with existing retail platform (App + Graphical User Interface)	Per store		13,000
	Per Trolley		\$30
Maintenance (10% of total cost)			
Total Cost (Approximate)			\$465

VI. FUTURE WORK AND RECOMMENDATION

It is reasonably justifiable that this application can further be improved in several aspects. The Geiger count based RSSI average is not a perfect measurement to ensure that the antenna only reads the tags inside the trolley. The Geiger count is calculated based on the distance between tag and antenna and the environmental conditions may also have an impact over the readings. Therefore, defining Geiger count % threshold will be very useful (as it defines for this application) to avoid reading tags far away from the shopping trolley boundaries.

Furthermore, a RF insulation for the shopping trolley can be suggested in order to avoid reading tags outside of the trolley. It may also need to encounter any potential coupling between antenna and trolley frame once the insulation is in place which needs further investigation. Furthermore, the accuracy of items in the trolley can be enhanced by scanning the items multiple times (eg: every 1second), which also provides more accurate real time information to the shopper.

Alternatively, the user interface can be further improved by providing shopping history which may aid shoppers to make

purchase decisions. Alternatively, the facility to download a pre-arranged shopping list is also useful for consumers. The development of “low cost” UHF RFID tag that can aid the reading in presence of metal (where Tag ID is difficult to read due to the interference generated from the radio frequency (RF) reflection from metal surfaces) and liquid (where RF energy gets absorbed due to continuous reorientation of polar molecules) are also a key future aspect to consider. Careful consideration of dielectric substrates (eg: thermoplastics) which may separate the antenna and ground plane and the use of printed folded dipole antenna (for a compact size) will be an area to explore in order to address a low cost UHF RFID label that may be used for the item level tagging. The implemented system can be further improved by the use of data mining techniques to gather product data at multiple points and also to identify consumer purchased patterns using neural networks.

VII. CONCLUSION

In this paper, the UHF RFID system has been successfully used for the smart shopping trolley application has been demonstrated. The items can be detected irrespective of its tag orientation, size and shape. These were the drawbacks mentioned in previous shopping trolley applications which were overcome in this application. The development of antenna and hybrid coupler is based on the original work which has been carried out by Monash Microwave, Antennas, RFID and Sensor laboratories for the development of partial discharge network in UHF frequencies. Finally, this particular application may bring novel experience for shoppers when they benefit from coordinated collaboration among technologies.

ACKNOWLEDGMENT

Special thanks goes to Monash Engineering workshop for their continuous support for the fabrication of antennas and preparing relevant jigs for the antenna frames as well as Mr Geoffrey Ramadan, the managing director of UMD (Unique Micro Design) for providing the hardware support needed for the application development.

REFERENCES

- [1] R. O'Neill. (21 June 2005, 21 June 2017). *Smart trolley shops for you*. Available: <http://www.smh.com.au/news/technology/checkout-chic-smart-trolley-shops-for-you/2005/07/20/1121539033473.html>
- [2] C. N. Megan Griffith-Greene / Marketplace. (28 Jan 2016, 22 June 2017). *Self Check Outs*. Available: <http://www.cbc.ca/news/business/marketplace-are-you-being-served-1.3422736>
- [3] ncr.com, "SELF-CHECKOUT: A GLOBAL CONSUMER PERSPECTIVE," ed: 3097 Satellite Boulevard . Duluth, Georgia 30096 . USA, 2014.
- [4] Foodlabels.industry.gov.au. (2017, 22 June 2017). *Australia's food labels are getting clearer*. Available: <http://www.foodlabels.industry.gov.au/>
- [5] prwire.com. (2005, 22 June 2017). *Fujitsu to Unveil Australia's First Intelligent Shopping Trolley at 2005 Retail Business Technology Expo* Available: <http://prwire.com.au/pr/2483/fujitsu-to-unveil-australias-first-intelligent-shopping-trolley-at-2005-retail-business-technology-expo>
- [6] amazon.com. (2017, 25 July). *AmazonGo*. Available: <https://www.amazon.com/b?node=16008589011>
- [7] R. Blundell. (20 May 2017). *Is Amazon Go shutting down the checkout for good? .* Available: <https://www.digitalpulse.pwc.com.au/amazon-go-strategy-retail-grocery/>
- [8] N. Garun. (12 Dec 2016, 22 June 2017). *Panasonic's smart shopping basket calculates your bill and bags your items*. Available: <https://www.theverge.com/2016/12/12/13920454/panasonic-smart-basket-automatically-bag-items-amazon-go>
- [9] S. L. F. DAILYMAIL.COM. (13 Dec 2016, 23 June 2017). *The smart shopping basket that can detect anything you put in it - and even automatically bag it for you*. Available: <http://www.dailymail.co.uk/sciencetech/article-4025982/The-smart-shopping-basket-detect-automatically-bag-you.html>
- [10] P. Z. Prof Nema Karmakar, Maneesha Kambukage "FPGA-CONTROLLED PHASED ARRAY ANTENNA DEVELOPMENT FOR UHF RFID READER," in *Handbook of Smart Antennas for RFID Systems*, P. N. Karmakar, Ed.: John Wiley & Sons, Inc., 2010.
- [11] S. Subedi, E. Pauls, and Y. D. Zhang, "Accurate Localization and Tracking of a Passive RFID Reader Based on RSSI Measurements," *IEEE Journal of Radio Frequency Identification*, vol. PP, no. 99, pp. 1-1, 2017.
- [12] D. M. Dobkin, "The RF in RFID: Passive UHF RFID in Practice,,"
- [13] F. Martinelli, "A Robot Localization System Combining RSSI and Phase Shift in UHF-RFID Signals," *IEEE Transactions on Control Systems Technology*, vol. 23, no. 5, pp. 1782-1796, 2015.
- [14] R. Arena. (2013, 15 November). *Applying RSSI filters for optimal RFID performance*. Available: <http://www.rfidarena.com/2013/8/28/applying-rssi-filters-for-optimal-rfid-performance.aspx>
- [15] GS1, "Regulatory status for using RFID in the EPC Gen2 (860 to 960 MHz) band of the UHF spectrum," ed, November 2016.
- [16] <https://www.microwaves101.com>. (2016, 12 November). *Branchline Couplers*. Available: <https://www.microwaves101.com/encyclopedias/branchline-couplers>
- [17] A. Sayeed, "Quadrature Branch-line Hybrid Coupler With Two Stages.," Department of Electronics & Telecommunication M.H.S.S. College of Engineering, Mumbai, India 2016.
- [18] R. Australia, "UHF RFID tag," ed, 2016.
- [19] Xerafy.com. (2012, 05 February). *Xerafy metal skin RFID label*. Available: <https://barcode.com/20120418884/euro-id-2012-showcases-xerafys-metal-skin-rfid-tags.html>
- [20] A. S. WRITERS. (2017, 05 November). *Australian shopping trolley dilemma*. Available: <http://www.ausfoodnews.com.au/2012/01/06/australian-shopping-trolley-dilemma-carrot-and-stick-strategies-for-recovery-of-the-ones-that-go-missing.html>
- [21] U. R. Reader. (2017, 15 November). *ID01 UHF RFID Reader*. Available: <https://core-electronics.com.au/id01-uhf-rfid-reader-usb.html>
- [22] C. Reports. (2017, 15 November). *What to Do when There Are Too Many Product Choices on the Store Shelves?* Available: <https://www.consumerreports.org/cro/magazine/2014/03/too-many-product-choices-in-supermarkets/index.htm>